Joint probability distribution

When we have two or more random variables, and we have reasons to believe that these variables are related, then we have to study them together. The probability distribution of two or more random variables is called a joint probability distribution.

We will study joint distribution of two random variables only.

Discrete Case:

Joint pmf

Let X and Y be two discrete random variables. The joint probability mass function or joint probability function or joint pmf of X and Y, denoted by $p(x, y)$, is defined as $p(x, y) = P(X = x, Y = y)$. It satisfies the following two conditions:

(i) $p(x, y) \geq 0$

(ii) $\sum_x \sum_y p(x, y) = 1$

Example:

Let $X =$ Number of houses, and $Y =$ Number of cars. Let the joint pmf of X and Y be given as follows:

x	y			
	0	1	2	3
0	0.30	0.15	0.10	0.05
1	0.10	0.10	0.05	0.05
2	0.01	0.03	0.04	0.02

Here,

$p(0, 1) = P(X = 0, Y = 1) = 0.15$,

$p(2, 3) = P(X = 2, Y = 3) = 0.02$,

and so on.

Marginal distributions

The (marginal) distribution of X, denoted by $p_X(x)$, is given by:

$$p_X(x) = P(X = x) = \sum_y p(x, y)$$

The (marginal) distribution of Y, denoted by $p_Y(y)$, is given by:

$$p_Y(y) = P(Y = y) = \sum_x p(x, y)$$

Exercise:

Determine the marginal distribution of X from the above example.

Solution:

x	0	1	2
$p_X(x)$	0.60	0.30	0.10

$p_X(1) = P(X = 1) = 0.30$

and so on.

Note:

Marginal distributions of X and Y are shown in the margins of the following table (in yellow and blue colors, respectively).

x	y				$p_X(x)$
	0	1	2	3	
0	0.30	0.15	0.10	0.05	0.60
1	0.10	0.10	0.05	0.05	0.30
2	0.01	0.03	0.04	0.02	0.10
$p_Y(y)$	0.41	0.28	0.19	0.12	1.00

Conditional distributions:

The conditional distribution of Y given $X = x$, denoted by $p_{Y|X}(y|x)$, is given by:

$$p_{Y|X}(y|x) = P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{p(x, y)}{p_X(x)}$$

Exercise:

For the above joint distribution, determine the conditional distribution of Y given $X = 1$.

Solution:

y	0	1	2	3
$p_{Y \mid X}(y \mid 1)$	1/3	1/3	1/6	1/6

Here,

$$p_{Y \mid X}(0 \mid 1) = \frac{0.10}{0.30} = 1/3$$

and so on.

Continuous Case: **Joint pdf**

Let X and Y be two continuous random variables. The joint probability density function or joint density or joint pdf of X and Y, denoted by $f(x, y)$, is a function that gives density at $X = x$ and $Y = y$. It satisfies the following two conditions:

(i) $f(x, y) \geq 0$

(ii) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dy\, dx = 1$

We can calculate probabilities by using the following formula:

$$P(a < X < b,\ \ c < Y < d) = \int_a^b \int_c^d f(x, y)\, dy\, dx$$

Example:

Check whether the following function is a joint pdf:

$$f(x, y) = \frac{1}{3}(x + y),\ \ 0 < x < 2,\ \ 0 < y < 1$$

Solution:

i)

$f(x,y) \geq 0$ for $0 < x < 2, 0 < y < 1.$

ii)

$$\int_0^2 \int_0^1 \frac{1}{3}(x+y)\,dy\,dx$$

$$= \int_0^2 \frac{1}{3}\left(xy + \frac{y^2}{2}\right)\Bigg|_0^1 dx$$

$$= \int_0^2 \frac{1}{3}\left(x + \frac{1}{2}\right) dx$$

$$= 1$$

Therefore, $f(x,y)$ is a joint pdf.

Exercise

Consider the following joint pdf:

$$f(x,y) = \frac{1}{3}(x+y), \quad 0 < x < 2, \quad 0 < y < 1.$$

Determine (a) $P(0.6 < X < 1.2,\ 0.2 < Y < 0.4)$ (b) $P(0.6 < X < 1.2)$ (c) density at $X = 0.5, Y = 0.4$ (d) $P(X = 0.5,\ Y = 0.4)$ (e) $P(X = 0.5, 0.3 < Y < 0.5)$.

Solution:

(a)

$$P(0.6 < X < 1.2,\ \ 0.2 < Y < 0.4)$$

$$= \int_{0.6}^{1.2} \int_{0.2}^{0.4} \frac{1}{3}(x+y)\,dy\,dx$$

$$= 0.048$$

(b)

$P(0.6 < X < 1.2)$

$= \int_{0.6}^{1.2} \int_{0}^{1} \frac{1}{3}(x + y)\, dy\, dx$

$= 0.28$

(c)

Density at $X = 0.5, Y = 0.4$

$= f(0.5, 0.4)$

$= \frac{1}{3}(0.5 + 0.4)$

$= 0.3$

(d)

$P(X = 0.5, Y = 0.4) = 0$

(e)

$P(X = 0.5, 0.3 < Y < 0.5) = 0$

Marginal distributions

The (marginal) distribution or density of X, denoted by $f_X(x)$, is given by:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$$

The (marginal) distribution or density of Y, denoted by $f_Y(y)$, is given by:

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$$

Exercise:

$f(x, y) = x + y, \;\; 0 < x < 1, \;\; 0 < y < 1$. Determine the marginal density of X.

Solution:

$f_X(x)$

$= \int_0^1 (x + y)\, dy$

$= x + 0.5, \ \ 0 < x < 1$

Conditional distributions:

The conditional distribution of Y given $X = x$, denoted by $f_{Y|X}(y|x)$, is given by:

$f_{Y|X}(y|x) = \dfrac{f(x, y)}{f_X(x)}$

Exercise:

For the above joint distribution, determine the conditional distribution of $f_{Y|X}(y|0.25)$.

Solution:

$f_X(x) = x + 0.5$

$\therefore f_X(0.25) = 0.25 + 0.5 = 0.75$

$f_{Y|X}(y|0.25)$

$= \dfrac{f(0.25, y)}{f_X(0.25)}$

$= \dfrac{y + 0.25}{0.75}$

$= \dfrac{1}{3}(4y + 1), \ \ 0 < y < 1$

Independence of two random variables

Discrete Case:

Let X and Y be two discrete random variables with joint distribution $p(x, y)$. X and Y are said to be independent if, for each pair of values x and y,

$P(X = x, Y = y) = P(X = x)\, P(Y = y)$.

That is, X and Y are said to be independent if

$p(x, y) = p_X(x)\, p_Y(y)$

for each pair of values x and y.

Example

Let the joint pmf of X and Y be as follows. Are X and Y independent?

x	y			
	0	1	2	3
0	0.24	0.18	0.12	0.06
1	0.12	0.09	0.06	0.03
2	0.04	0.03	0.02	0.01

Solution

x	y				$p_X(x)$
	0	1	2	3	
0	0.24	0.18	0.12	0.06	0.60
1	0.12	0.09	0.06	0.03	0.30
2	0.04	0.03	0.02	0.01	0.10
$p_Y(y)$	0.40	0.30	0.20	0.10	1.00

$p_X(0)\, p_Y(0) = 0.60 \times 0.40 = 0.24 = p(0, 0)$

It is true for each other pair of values of X and Y.

Therefore, X and Y are independent.

Example

Let the joint pmf of X and Y be as follows. Are X and Y independent?

x	y			
	0	1	2	3
0	0.24	0.18	0.12	0.06
1	0.12	0.09	0.07	0.02
2	0.04	0.03	0.01	0.02

Solution

x	y				$p_X(x)$
	0	1	2	3	
0	0.24	0.18	0.12	0.06	0.60
1	0.12	0.09	0.07	0.02	0.30
2	0.04	0.03	0.01	0.02	0.10
$p_Y(y)$	0.40	0.30	0.20	0.10	1.00

$p_X(1)\, p_Y(2) = 0.30 \times 0.20 = 0.06 \neq p(1, 2)$

Therefore, X and Y are NOT independent.

Continuous Case:

Let X and Y be two continuous random variables. X and Y are independent if

$f(x, y) = f_X(x)\, f_Y(y)$

for all x and y.

Example:

The joint pdf of X and Y is given below. Are X and Y independent?

$f(x, y) = 4xy, \;\; 0 < x < 1, \;\; 0 < y < 1$

Solution:

$f_X(x)$

$$= \int_0^1 4xy\, dy$$

$= 2x, \;\; 0 < x < 1$

$f_Y(y)$

$$= \int_0^1 4xy\, dx$$

$= 2y, \;\; 0 < y < 1$

$f_X(x)\, f_Y(y) = 2x\; 2y = 4xy = f(x, y)$

Therefore, X and Y are independent.

Example

The joint pdf of X and Y is given below. Are X and Y independent?

$f(x, y) = x + y, \;\; 0 < x < 1, \;\; 0 < y < 1.$

Solution

$f_X(x)$

$$= \int_0^1 (x + y)\, dy$$

$= x + 0.5, \;\; 0 < x < 1$

$f_Y(y)$

$$= \int_0^1 (x + y)\, dx$$

$= y + 0.5, \;\; 0 < y < 1$

$f_X(x)\, f_Y(y) = (x + 0.5)(y + 0.5) \neq f(x, y)$

Therefore, X and Y are NOT independent.

Mathematical Expectation

Discrete Case:

Let X be a discrete random variable with pmf $p(x)$. The mathematical expectation or expected value or population mean of X, denoted by $E(X)$ or μ, is defined as

$$\mu = E(X) = \sum_x x\, p(x)$$

Example

Let X have the following pmf.

x	0	1	4
$p(x)$	0.25	0.50	0.25

$$\mu = \sum_x x\, p(x)$$

$$= 0 \times 0.25 + 1 \times 0.50 + 4 \times 0.25$$

$$= 1.5$$

Comparison with sample mean

Sample mean is defined as

$$\bar{x} = \frac{1}{n} \sum_i x_i\, f_i = \sum_i x_i \frac{f_i}{n}$$

Population mean is defined as

$$\mu = \sum_x x\, p(x)$$

That is, for sample mean each value is multiplied by relative frequency, while for population mean each value is multiplied by probability.

Example: population mean vs sample mean

Consider the following population (pmf):

x	1	2	7
$p(x)$	0.25	0.50	0.25

Here, the population mean is

$$\mu = \sum_x x\, p(x)$$

$$= 1 \times 0.25 + 2 \times 0.50 + 7 \times 0.25$$

$$= 3.00$$

Let us take a sample of size $n = 1000$ from this population and construct the following frequency table.

X	Frequency	Relative Frequency
1	240	0.24
2	510	0.51
7	250	0.25

The sample mean is

$$\bar{x} = \sum_i x_i \frac{f_i}{n}$$

$$= 1 \times 0.24 + 2 \times 0.51 + 7 \times 0.25$$

$$= 3.01$$

Continuous Case:

Let X be a continuous random variable with pdf $p(x)$. The mathematical expectation or expected value or population mean of X, denoted by $E(X)$ or μ, is defined as

$$\mu = E(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

Example

Let X have the following pdf.

$$f(x) = \frac{1}{9}x^2, \;\; 0 < x < 3.$$

$$\mu = E(X) = \int_0^3 x\, \frac{1}{9}x^2\, dx = 2.25$$

Example: population mean vs sample mean

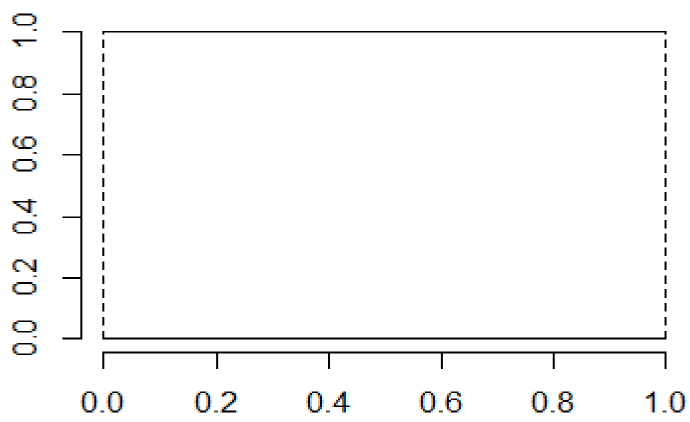
Let us consider the population (pdf):

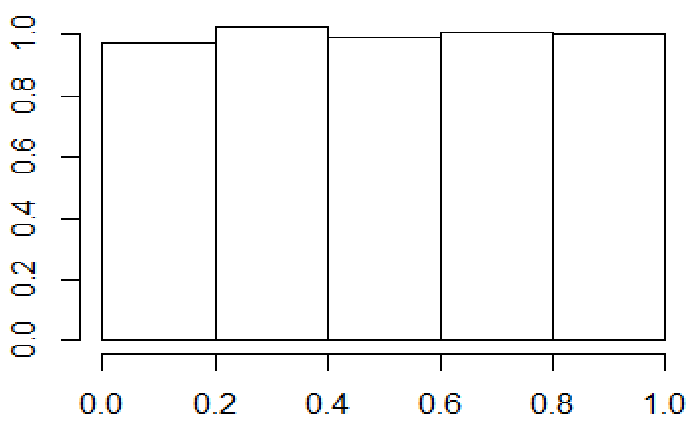
$f(x) = 1, \;\; 0 < x < 1$

Let us take a sample of size $n = 1000$ from this population and construct the following frequency table.

X	Frequency	Relative Frequency
0.0 – 0.2	196	0.196
0.2 – 0.4	204	0.204
0.4 – 0.6	199	0.199
0.6 – 0.8	201	0.201
0.8 – 1.0	200	0.200

Let us compare the pdf (first plot below) and histogram (second plot below):





The population mean is

$$\mu = E(X) = \int_0^1 x \, 1 \, dx = 0.50$$

The sample mean is

$$\bar{x} = \sum_i m_i \frac{f_i}{n}$$

$$= 0.1 \times 0.196 + 0.3 \times 0.204 + 0.5 \times 0.199 + 0.7 \times 0.201 + 0.9 \times 0.200$$

$$= 0.501$$

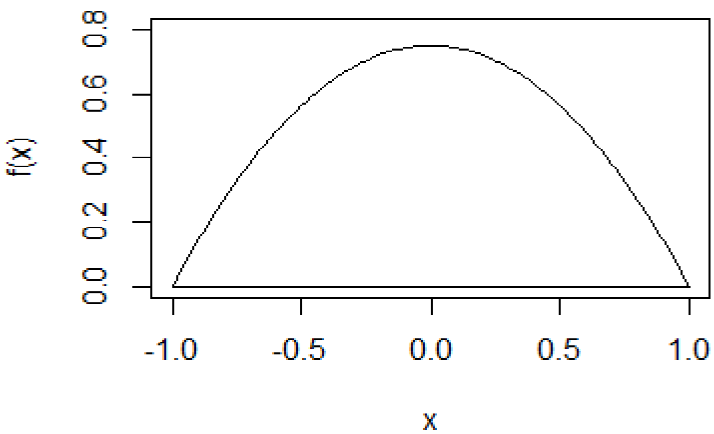
Example

Let X have the following pdf.

$$f(x) = 0.75\,(1 - x^2), \quad -1 < x < 1.$$

$$\mu = E(X) = \int_{-1}^{1} x \, 0.75\,(1 - x^2)\, dx = 0$$

This is a symmetric distribution. So, mean is exactly in the middle.



Exercise

Let X have the following pdf. Calculate the mean of X.

$$f(x) = x + 0.5, \quad 0 < x < 1.$$

Solution

Do it yourself.

Expected value of function of random variable

Discrete Case:

Let X be a discrete random variable with pmf $p(x)$. The expected value or mean of $Y = g(X)$ is obtained as

$$\mu_Y = E(g(X)) = \sum_x g(x)\, p(x)$$

Example

Let X have the following pmf. Obtain the mean of $Y = X^2$.

x	0	1	2
$p(x)$	0.35	0.40	0.25

$$E(X^2) = \sum_x x^2\, p(x)$$

$$= 0^2 \times 0.35 + 1^2 \times 0.40 + 2^2 \times 0.25$$

$$= 1.40$$

Example

Let X have the following pmf. Obtain the variance of $Y = X^2$.

x	-1	0	1	2
$p_X(x)$	0.10	0.50	0.20	0.20

$$E(X^2) = \sum_x x^2\, p(x)$$

$$= (-1)^2 \times 0.10 + 0^2 \times 0.50 + 1^2 \times 0.20 + 2^2 \times 0.20$$

$$= 1.10$$

Continuous Case:

Let X be a continuous random variable with pdf $f(x)$. The expected value or mean of $Y = g(X)$ is obtained as

$$\mu_Y = E(g(X)) = \int_{-\infty}^{\infty} g(x)\, f(x)\, dx$$

Example

Let X have the following pdf. Obtain the mean of $Y = X^2$.

$f(x) = \frac{1}{9}x^2, \;\; 0 < x < 3.$

$\mu_Y = E(X^2) = \int_0^3 x^2 \, \frac{1}{9} x^2 \, dx = 5.40$

Properties of expectation

1. $E(c) = c$
2. $E(aX + b) = a\,E(X) + b$
3. $E(X + Y) = E(X) + E(Y)$

Variance

Variance of a random variable X, denoted by σ^2 or $V(X)$, is defined as

$\sigma^2 = V(X) = E((X - E(X))^2) = E(X^2) - (E(X))^2$ (after simplification)

Example (discrete)

Let X have the following pmf. Obtain the variance of X.

x	0	1	2
$p(x)$	0.35	0.40	0.25

$\mu = \sum_x x\,p(x)$

$= 0 \times 0.35 + 1 \times 0.40 + 2 \times 0.25$

$= 0.90$

$E(X^2) = \sum_x x^2\,p(x)$

$= 0^2 \times 0.35 + 1^2 \times 0.40 + 2^2 \times 0.25$

$= 1.40$

$V(X) = E(X^2) - \mu^2 = 1.40 - 0.90^2 = 0.59$

Example (continuous)

Let X have the following pdf. Obtain the variance of X.

$f(x) = \frac{1}{9}x^2, \;\; 0 < x < 3.$

$\mu = \int_0^3 x \; \frac{1}{9}x^2 \, dx = 2.25$

$E(X^2) = \int_0^3 x^2 \; \frac{1}{9}x^2 \, dx = 5.40$

$V(X) = E(X^2) - \mu^2 = 5.40 - 2.25^2 = 0.3375$

Exercise

Let X have the following pdf. Calculate the variance of X.

$f(x) = x + 0.5, \;\; 0 < x < 1.$

Solution

Do it yourself.

Properties of variance

1. $V(c) = 0$
2. $V(aX + b) = a^2V(X)$